

All-gap endpoint singularity persists under insertion

Asymptotic diagnostic, not a size optimization

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Let G_A be the reduced free nilpotent group on an alphabet A , and let X_i denote its degree-one generators. For a positive word W , put

$$S(W) = \text{span}\{\text{Ad}_{P_q} X_i : i \in A, q \text{ a gap of } W\} \subseteq \mathfrak{g}_A,$$

where P_q is the signature of the prefix ending at the gap. This is the right-trivialized differential of inserting arbitrary infinitesimal letter weights at all gaps. Write $H = \sum_{i \in A} X_i$.

Lemma 1. *If W is fair with signature $\exp(MH)$, then*

$$S(W^t) = \sum_{j=0}^{t-1} e^{jM \text{ad } H} S(W).$$

Consequently, if $S(W)$ is invariant under $\text{ad } H$, then $S(W^t) = S(W)$ for every positive integer t .

Proof. Every prefix of W^t is the signature of some number j of complete copies, followed by a prefix P of one copy. Its adjoint action is $\text{Ad}_{\exp(jMH)P} = e^{jM \text{ad } H} \text{Ad}_P$. The gap at the final endpoint contributes no new vectors, since it is already a gap of the last copy. Invariance gives the last assertion; the first copy supplies the reverse inclusion. $\square$

Lemma 2. *Let V be a word on $B \supseteq A$ whose restriction to A is W^t . Under the natural Lie-algebra projection $\pi : \mathfrak{g}_B \rightarrow \mathfrak{g}_A$, one has*

$$\pi(S(V)) \subseteq S(W^t).$$

In particular, if $S(W^t) \neq \mathfrak{g}_A$, then $S(V) \neq \mathfrak{g}_B$.

Proof. Projection commutes with the adjoint action. An insertion of a letter outside A projects to zero. An insertion of a letter of A projects to its insertion at the corresponding gap of the restricted word. A surjective ambient tangent space would have surjective projection. $\square$

Corollary 3. *There are positive fair words with singular all-gap endpoint maps on arbitrarily large alphabets.*

Proof. An exact rational-arithmetic diagnostic supplies one fair word W with $\dim S(W) = 86 < 89 = \dim \mathfrak{g}_A$ and $[H, S(W)] \subseteq S(W)$. The files `certify_all_gap_singularity.py` and `certify_all_gap_power_singularity.py` verify these assertions by integer matrix factorizations through an invertible 86×86 minor, independently of floating-point arithmetic. The input is the published fair word stored in `../equal_size/meyer.2023_five_dice_sixty.json`. Apply the standard one-letter insertion construction repeatedly. At each stage, restriction to the previous alphabet is a positive integer power of the preceding word. Hence restriction to A remains a power of W . The two lemmas give singularity at every stage. $\square$

Corollary 4 (Fairness does not force truncated surjectivity). *For every $r \geq 5$ and every $n \geq 2r$, there is a fully permutation-fair positive word on n letters whose all-gap endpoint map is not surjective even onto the reduced Lie algebra truncated through degree r . In particular, fairness through degree $2r$ does not imply full endpoint regularity through degree r .*

Proof. Use the same inherited construction on n letters. Projection to its original five-letter alphabet factors through truncation at degree r , since that entire reduced Lie algebra has degree at most five. Surjectivity of the truncated tangent map would therefore surject onto the original five-letter algebra, contradicting the inherited singularity. Full fairness implies fairness through $2r$. The actual face-weight differential is a subspace of the all-gap insertion differential, so it cannot restore this surjectivity. $\square$

Scope. This closes the proposed route that every positive fair word must have a regular all-gap endpoint map, even after repetition, and also the stronger-accuracy-to-lower-degree surjectivity variant. It gives no lower bound on the relative corank and does not exclude a weaker factorial lower bound for the rank. The fixed input is used solely to falsify a universal asymptotic lemma; no small-size optimization is being pursued.